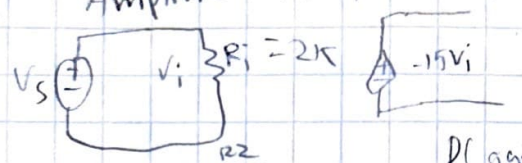
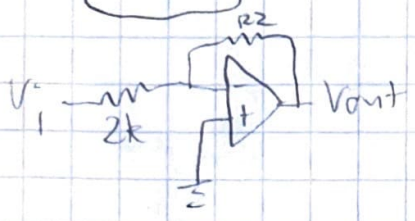
Dylan Thornburg EEN 115 Prelab 3

1a. Amplifier model when signal is small



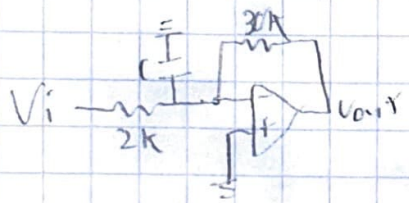


DC gain = +15 for inverting amp

$-15 = \frac{-R2}{R_i}$; $15 = \frac{R2}{2k}$

$R2 = 30k$

Now we have to give it low pass ability.



This will work b/c when $\omega \approx 0$ (will be open and the circuit will act like an amp.

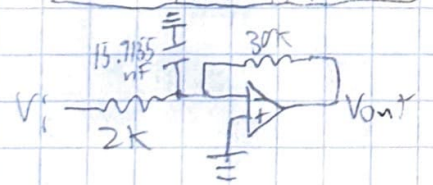
When $\omega = \infty$ (will be a wire straight to ground which will force Vout to be 0 so gain $\frac{V_{out}}{V_{in}}$ then $= 0$.

To find C, we must recall the 3dB (cutoff) freq is 5kHz, let's solve for C

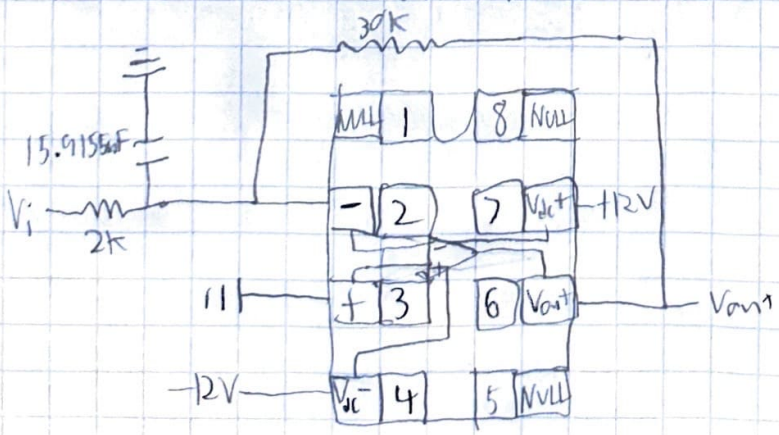
$\frac{1}{2\pi RC}$ = cutoff freq ; $\frac{1}{2\pi 2k \cdot C} = 5k$

$\frac{1}{2\pi 2k \cdot 5k} = C$; $C = 15.9155$ nF or ≈ 16 nF

Final circuit



1b. 741 OP AMP IMPLEMENTATION



1c. Effect of Amplifier

i.

$$V_{in}(t) = \frac{1}{2} + \frac{1}{10}\sin(1000\pi t)$$

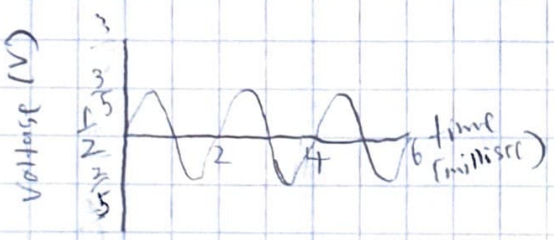
Frequencies below 6kHz?
Yes. All pass

$$\frac{1}{2}\cdot -15 + -15\cdot\frac{1}{10}\sin(1000\pi t)$$

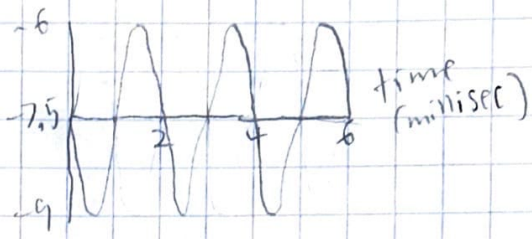
$$V_{out}(t) = -\frac{15}{2} - \frac{3}{2}\sin(1000\pi t)$$

Does clipping happen? No
$\frac{15}{2} + \frac{3}{2} < 12$

$V_{in}(t)$



$V_{out}(t)$



ii.

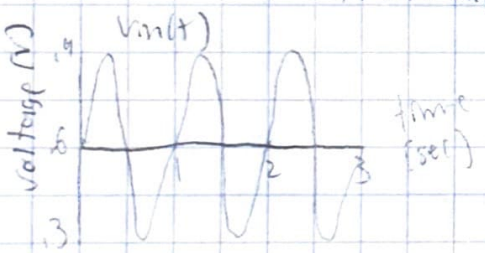
$V_{in} = .6 + .3\sin(2\pi t)$, all freq will pass

$-15\cdot .6 = -9 \quad -15\cdot .3 = -4.5$

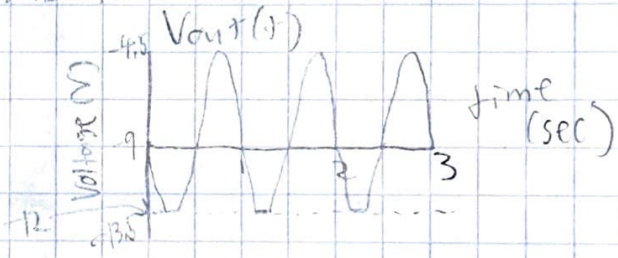
$V_{out} = -9 - 4.5\sin(2\pi t)$

It will clip as $|-13.5| > |\pm 12|$

$V_{in}(t)$



$V_{out}(t)$



iii.

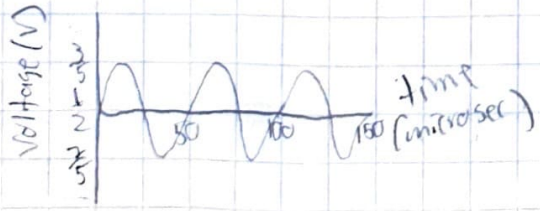
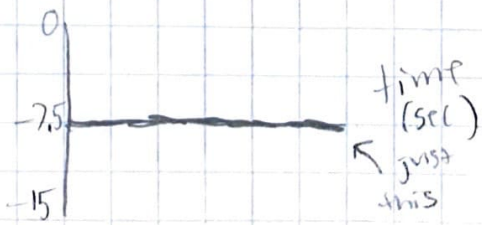
$$V_{in} = \frac{1}{2} + \frac{1}{10}\sin(40000\pi t)$$

$\frac{40000\pi}{2\pi} = 20000 \quad 20000 > 5000$
so the sin function won't be passed

$$V_{out} = \frac{1}{2}\cdot -15 + \cancel{-15\cdot\frac{1}{10}\sin(40000\pi t)}$$

$V_{out} = -7.5$ no clipping here

$V_{in}(t)$



$V_{out}(t)$